

Teddy Bear Wheelchair Team Final Design Report

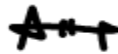
Section 2 Team 01, Netherlands

Authors: Aly Anany (1263531), Adnaan Dewan (1283018), Avneet Aujla (1268674), Alexander Falkiner (1280499), Aryan Ashraf (1286850), Anthony Gorin (1303928)

Date: November 26, 2023

Prepared for: University of Guelph Engineering Grading Committee

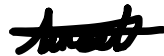
Aly Anany



Adnana Dewan



Avneet Aujla



Alexander Falkiner



Aryan Ashraf



Anthony Gorin



The authors take shared credit and responsibility for the content within this report.

Table of Contents

1. TBWC Design Summary	3
1.1 Introduction	3
1.2 Photos of TBWC.....	3
.....	4
.....	4
1.3 Bill of Materials	5
1.4 Cost analysis Tables.....	6
1.4.1 Final TBWC Production Cost- Cost of a Single TBWC Unit.....	6
1.4.2 Development cost- SOE Supplied Materials	7
1.4.3 Development cost- Additional Purchased or sourced Materials.....	8
1.5 Aesthetics discussion	8
1.6 Design Process Summary.....	9
2. TBWC Performance Summary.....	9
3. Reflection.....	11
3.1 Dribble Ramp (Speed)	11
3.2 Free Kick Shooting (Accuracy, Consistency)	12
3.3 Mass.....	13
3.4 Aesthetics	13
3.5 Safety	14
3.6 Cost Analysis Table	15
3.7 Life Cycle Analysis	16
4. Reflection- Team Dynamics.....	16
Engineering Analysis A- Centre of Mass Determination and Tipping Analysis	17
A.1 Engineering Analysis A.....	17
Engineering Analysis B- Life Cycle Impacts Analysis.....	18
B.1 Engineering Analysis B.....	18
Engineering Analysis C- Arduino Code.....	19
C.1 Engineering Analysis C.....	19
Engineering Analysis D- Isometric Drawing of Final TBWC.....	21
D.1 Engineering Analysis D	21
Engineering Analysis E- Orthographic Projections of Final TBWC.....	22
E.1 Engineering Analysis E	22
Engineering Analysis F- TBWC Electrical Circuit	25

F.1 Engineering Analysis F.....	25
<i>Appendix G- TBWC Aesthetic Assessment Rubric and Grading</i>	26
<i>Appendix H- TBWC Week 7 Revised Team Contract.....</i>	27

1. TBWC Design Summary

1.1 Introduction

The primary goal of the Netherlands team was to reduce the weight of the TBWC as early in the design phase as possible. To accomplish this, we worked hard to limit the design to a single motor, which made it possible for us to create a launching mechanism that was solely dependent on the movement of the chassis. Furthermore, the TBWC chassis went under numerous modifications closing the final performance demonstration, from several changes to launcher placement, and switching the wheel foundation from 4 to 3 wheels to effectively optimize speed and weight. The purpose of the following report is to provide a thorough assessment of the overall TBWC performance and development. This will include detailed tables outlining materials as well as the anticipated costs for clients to produce the following TBWC. In addition, it will include visual components that will showcase the TBWC's completed subsystems and the aesthetic aspects that reflect the country of Netherlands alongside discussions to give insight on the team's decision and results.

1.2 Photos of TBWC



Figure 1.1 Isometric View of Chassis & Launching Mechanism

The following image displays the chassis and the slightly elevated launching mechanism as whole. The narrow front axle with one wheel and the wide rear axle with two wheels aiding in stability.

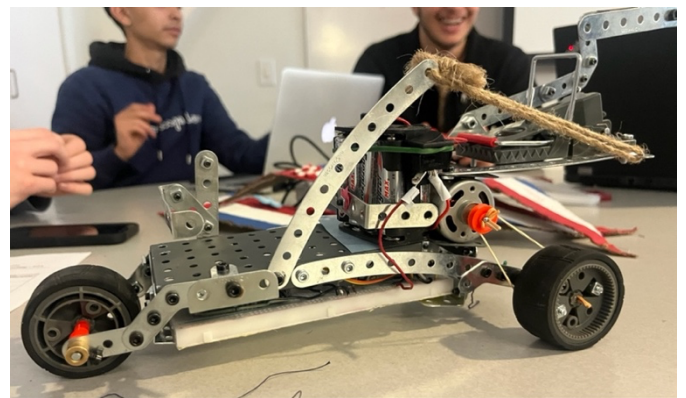


Figure 1.2 Front View of Chassis & Launching Mechanism

The following image displays the organization of the chassis with wires being tucked away and every functional component secured.

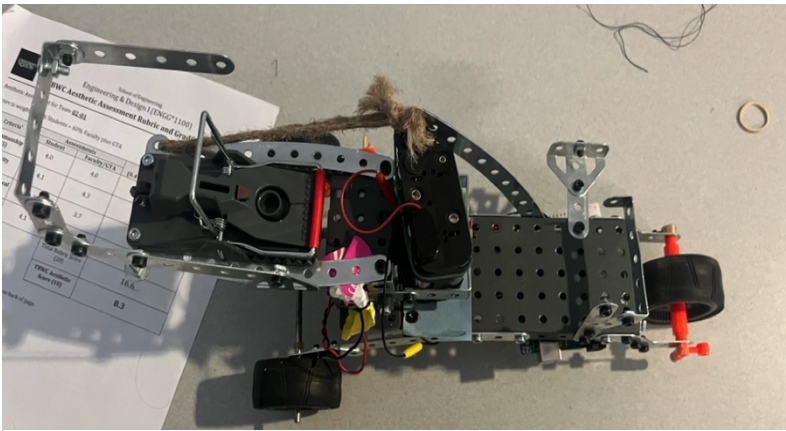


Figure 1.3 Top View of Chassis & Launching Mechanism

The following image displays the launcher being angled to the left side to avoid any interference with the teddy bear and to improve accuracy of the launcher. Additionally shows the component that secures the Teddy Bear

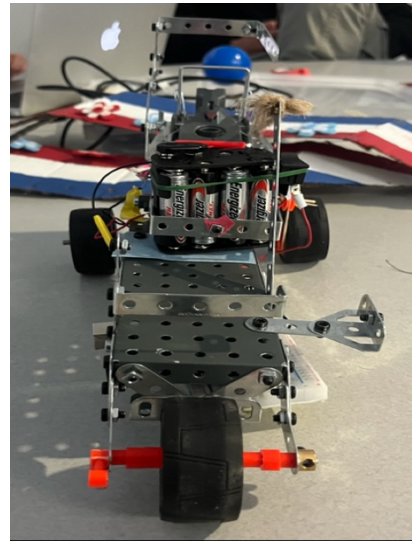


Figure 1.4 Front View of Chassis & Launching Mechanism

The following image displays the slight tilt in the foundation which improves the accuracy of the launcher.

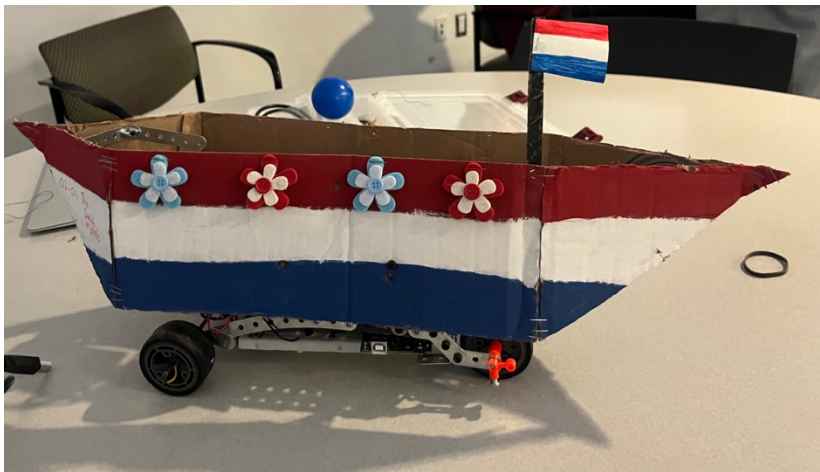


Figure 1.5 Rolling Chassis with Aesthetic Component

The following image displays the shell of the chassis incorporating the colors of Netherlands and has flowers to represent the flower fields.

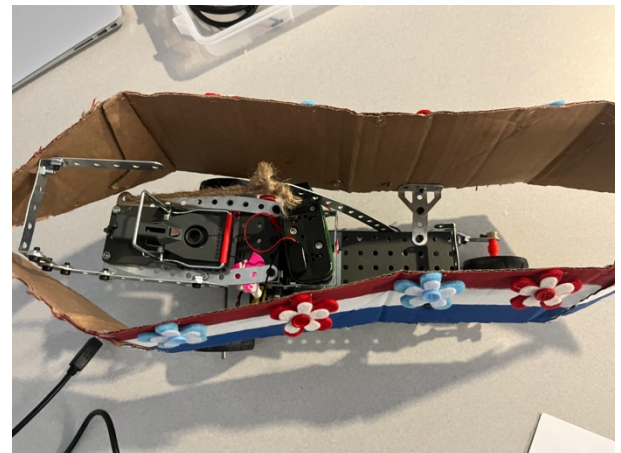


Figure 1.6 Top of Aesthetic Component

The following image displays the height of shell that encloses the Teddy Bear from falling out.

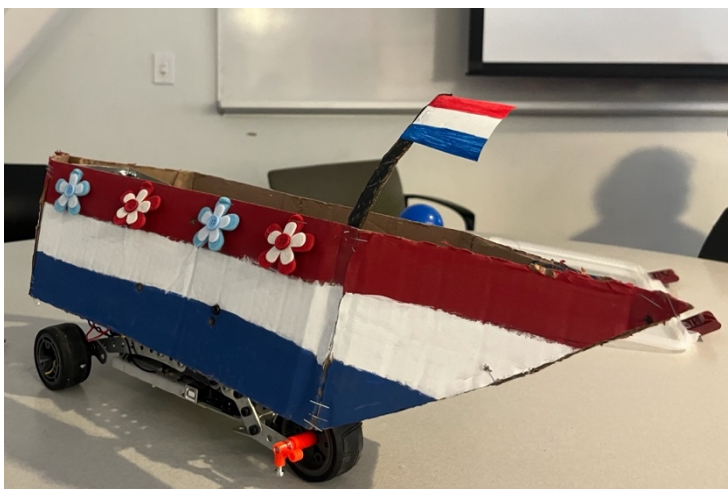


Figure 1.7 Isometric View of Aesthetic Component

The following image displays the shell of chassis with the flag standing tall and each aesthetic component on display.

1.3 Bill of Materials

Material & Part Number	Quantity	Unit	Part Description
12V/8-AA Battery Pack	1	Indiv.	Power Source
Battery Pack Clip	1	Indiv.	Battery Holder
AA Batteries	8	Indiv.	Power Supply
DC Brushed Motor	1	Indiv.	Motor that spins a pulley allowing for movement
Breadboard	1	Indiv.	For the circuit and allows code to function
Solid Core Wire (1 m)	1	Indiv.	Movement of
Arduino UNO	1	Indiv.	Brain of the system instructs the motor
L6205 Motor Driver Integrated Circuit	1	Indiv.	
5.6 nF Ceramic capacitor (562)	1	Indiv.	Stores electrical energy
100uF Electrolytic Capacitor (100uF)	1	Indiv.	Stores electrical energy
100nF Ceramic Capacitor (104)	1	Indiv.	Stores electrical energy
100 Ohm Resistor	1	Indiv.	Protects circuit from receiving too much voltage
10nF Ceramic Capacitor (103)	1	Indiv.	Stores electrical energy
1N4148 Diode	2	Indiv.	Allows current to flow
220nF Ceramic Capacitor (224)	1	Indiv.	Stores electrical energy
100 kOhm Resistor	1	Indiv.	Protects circuit from receiving too much voltage
Meccano Part 52 (87 Holes)	1	Part	Foundational Base of the chassis
Meccano Part 51E (21 Holes)	2	Part	Motor Holder
Meccano Part 89 (11 Holes)	5	Part	Suspension tool/ launcher holder
Meccano Part 9F (6 Holes)	2	Part	Breadboard Holder
Meccano Part 133C (4 Holes)	7	Part	Supports the axles
Meccano Part 51F (25 Holes)	1	Part	Battery Holder
Meccano Part 161 (12 Holes)	2	Part	Breadboard Holder
Meccano Part 47 (11 Holes)	1	Part	Battery Pack Holder
Meccano Part 316B (5 inches)	1	Part	Rear wheel axle
Meccano Part 17 (3 inches)	1	Part	Front wheel axle
Meccano Nuts		Part	Secures parts
Meccano Bolts		Part	Secures parts
Meccano Spacers		Part	Helps prevent axle from shifting
Meccano Part 22B3P	1	Part	Rotates rear axle
Meccano Part 23B	1	Part	Connected to motor creating movement
Meccano Part 187C	3	Part	Wheel that spins with the axle allowing movement
Meccano Part 142R	3	Part	Tire that creates traction
On-Off Switch	1	Indiv.	On-Off Switch
Meccano Part 5 (5 Holes)	2	Part	Teddy Bear Holder/ Secures Launching Mechanism
Meccano Part 6A (3 Holes)	3	Part	Shell Reinforcement
Meccano Part 48A (7 Holes)	1	Part	Teddy Bear Holder
Meccano Part 126 (7 Holes)	2	Part	Secures Launching Mechanism
Meccano Part 4 (6 Holes)	2	Part	Secures Launching Mechanism
Cardboard Box	70	g	Shell of Chassis
Roll of String	20	g	Launching Mechanism Control
Red Paint by Craft	1	2 oz	Visual Component
White Paint by Craft	1	2 oz	Visual Component
Blue Paint by Craft	1	2 oz	Visual Component
Flower Cutouts	15	g	Visual Component
Gold & Natural Brown Twine	1	75 ft.	Secures Launching Mechanism
Plastic Mosustrap (6-Pack)	1	pkg	Launcher

1.4 Cost analysis Tables

1.4.1 Final TBWC Production Cost- Cost of a Single TBWC Unit

Final TBWC Production Cost Analysis Table

Development Cost	Total Cost (\$)	Single TBWC Unit cost (\$)
SOE Supplied Materials	\$ 146.80	\$ 146.80
Purchased or Sourced Materials	\$ 27.05	\$ 27.05
Total		\$ 173.85

In total it costs \$173.85 to produce a single unit of the following TBWC. Considering the optimization of weight there are fewer parts than other competitor TBWC designs, in result it will reduce the cost for production of a singular unit. The following two tables will go into more detail of each individual part used for development including SOE supplied materials and purchased or sourced materials. Parts/materials bolded in a faint red are to represent parts used during the development process but later, replaced or completely taken out of the final design.

1.4.2 Development cost- SOE Supplied Materials

TBWC Development Cost- SOE Supplied Materials Cost Analysis Table

Date	Material & Part Number	Quantity	Unit	Cost/ Part (\$)	Cost (\$)	Cost Source
2023-10-18	12V/8-AA Battery Pack	1	Indiv.	\$ 5.00	\$ 5.00	TBWC Project Document
	Battery Pack Clip	1	Indiv.	\$ 3.00	\$ 3.00	TBWC Project Document
	AA Batteries	8	Indiv.	\$ 1.00	\$ 8.00	TBWC Project Document
	DC Brushed Motor	1	Indiv.	\$ 10.00	\$ 10.00	TBWC Project Document
	Breadboard	1	Indiv.	\$ 5.00	\$ 5.00	TBWC Project Document
	Solid Core Wire (1 m)	1	Indiv.	\$ 2.00	\$ 2.00	TBWC Project Document
	Arduino UNO	1	Indiv.	\$ 40.00	\$ 40.00	TBWC Project Document
	L6205 Motor Driver Integrated Circuit	1	Indiv.	\$ 12.00	\$ 12.00	TBWC Project Document
	5.6 nF Ceramic capacitor (562)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100uF Electrolytic Capacitor (100uF)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100nF Ceramic Capacitor (104)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100 Ohm Resistor	1	Indiv.	\$ 0.10	\$ 0.10	TBWC Project Document
	10nF Ceramic Capacitor (103)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	1N4148 Diode	2	Indiv.	\$ 0.50	\$ 1.00	TBWC Project Document
	220nF Ceramic Capacitor (224)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100 kOhm Resistor	1	Indiv.	\$ 0.10	\$ 0.10	TBWC Project Document
	Meccano Part 52 (87 Holes)	1	Part	\$ 8.70	\$ 8.70	TBWC Project Document
	Meccano Part 51E (21 Holes)	2	Part	\$ 2.10	\$ 4.20	TBWC Project Document
	Meccano Part 89 (11 Holes)	2	Part	\$ 1.10	\$ 2.20	TBWC Project Document
	Meccano Part 9F (6 Holes)	2	Part	\$ 0.60	\$ 1.20	TBWC Project Document
	Meccano Part 133C (4 Holes)	6	Part	\$ 0.40	\$ 2.40	TBWC Project Document
	Meccano Part 51F (25 Holes)	1	Part	\$ 2.50	\$ 2.50	TBWC Project Document
	Meccano Part 161 (12 Holes)	2	Part	\$ 1.20	\$ 2.40	TBWC Project Document
	Meccano Part 47 (11 Holes)	1	Part	\$ 1.10	\$ 1.10	TBWC Project Document
	Meccano Part 316B (5 inches)	1	Part	\$ 5.00	\$ 5.00	TBWC Project Document
	Meccano Part 17 (3 inches)	1	Part	\$ 3.00	\$ 3.00	TBWC Project Document
	Meccano Nuts		Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Bolts		Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Spacers		Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Part 22B3P	1	Part	\$ 1.00	\$ 1.00	TBWC Project Document
	Meccano Part 23B	1	Part	\$ 0.50	\$ 0.50	TBWC Project Document
	Meccano Part 187C	4	Part	\$ 1.00	\$ 4.00	TBWC Project Document
	Meccano Part 142R	4	Part	\$ 1.00	\$ 4.00	TBWC Project Document
2023-10-23	Meccano Part 51F (25 Holes)	1	Part	\$ 2.50	\$ 2.50	TBWC Project Document
	Meccano Part 6 (5 Holes)	2	Part	\$ 0.50	\$ 1.00	TBWC Project Document
	Meccano Part 126 (7 Holes)	2	Part	\$ 0.70	\$ 1.40	TBWC Project Document
	Meccano Part 1B (15 Holes)	1	Part	\$ 1.50	\$ 1.50	TBWC Project Document
	Meccano Part 4 (6 Holes)	2	Part	\$ 0.60	\$ 1.20	TBWC Project Document
	Meccano Part 51G (15 Holes)	3	Part	\$ 1.50	\$ 4.50	TBWC Project Document
	Meccano Part 51C (6 Holes)	1	Part	\$ 0.60	\$ 0.60	TBWC Project Document
	Meccano Part 17 (3 inches)	1	Part	\$ 3.00	\$ 3.00	TBWC Project Document
	Meccano Part 21P3P	1	Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Part 187C	3	Part	\$ 1.00	\$ 3.00	TBWC Project Document
	Meccano Part 142R	3	Part	\$ 1.00	\$ 3.00	TBWC Project Document
	On-Off Switch	1	Indiv.	\$ 3.00	\$ 3.00	TBWC Project Document
2023-11-06	Meccano Part 5 (5 Holes)	1	Part	\$ 0.50	\$ 0.50	TBWC Project Document
	Meccano Part 6A (3 Holes)	1	Part	\$ 0.30	\$ 0.30	TBWC Project Document
2023-11-15	Meccano Part 48A (7 Holes)	1	Part	\$ 0.70	\$ 0.70	TBWC Project Document
	Meccano Part 89 (11 Holes)	3	Part	\$ 1.10	\$ 3.30	TBWC Project Document
	Meccano Part 126 (7 Holes)	2	Part	\$ 0.70	\$ 1.40	TBWC Project Document
	Meccano Part 4 (6 Holes)	2	Part	\$ 0.60	\$ 1.20	TBWC Project Document
	Meccano Part 5 (5 Holes)	1	Part	\$ 0.50	\$ 0.50	TBWC Project Document
	Meccano Part 133C (4 Holes)	1	Part	\$ 0.40	\$ 0.40	TBWC Project Document
	Meccano Part 6A (3 Holes)	2	Part	\$ 0.30	\$ 0.60	TBWC Project Document
	Total Cost				\$ 146.80	

1.4.3 Development cost- Additional Purchased or sourced Materials

TBWC Development Cost- Team Purchased or Sourced Material Cost Analysis Table

Date	Material & Part Number	Quantity	Unit	Cost/ Part (\$)	Cost (\$)	Cost Source
2023-11-08	Cardboard Box	70	g	\$ 0.01	\$ 0.70	Recyclable material cost- TBWC Project Document
	Roll of String	20	g	\$ 0.20	\$ 0.20	Recyclable material cost- TBWC Project Document
	Red Paint by Craft	1	2 oz	\$ 1.00	\$ 1.00	https://www.michaels.com/product/acrylic-paint-by-craft-smart-2oz-10694310?michaelStore=5184&inv=8
	White Paint by Craft	1	2 oz	\$ 1.00	\$ 1.00	https://www.michaels.com/product/acrylic-paint-by-craft-smart-2oz-10694310?michaelStore=5184&inv=8
	Blue Paint by Craft	1	2 oz	\$ 1.00	\$ 1.00	https://www.michaels.com/product/acrylic-paint-by-craft-smart-2oz-10694310?michaelStore=5184&inv=8
	Flower Cutouts	15	g	\$ 0.15	\$ 0.15	Recyclable material cost- TBWC Project Document
	Gold & Natural Brown Twine	1	75 ft.	\$ 3.00	\$ 3.00	natural-brown-twine-by-ashland-10721856?michaelStore=5184&inv=98
2023-11-13	Wooden Mousetrap (2-Pack)	1	pkg	\$ 2.28	\$ 2.28	https://www.walmart.ca/en/ip/tomcat-mouse-wood-traps-2-pack-not-applicable/6090205841408?skuid=6090205841409&offerId=6090205841409&region-id=202200&empid=SEM-CA-33350-CCLODIMZMM&utm-id=SEM-CA-33350-CCLODIMZMM&utm-medium=paid-search&utm-source=google&utm_campaign=always-on&gad_source=4&gclid=CjwKCAiA9ourBhAVEiwA3L5RFrnSEcGTC4YThwqZlpeFolk1X86xRhmeal-A4vGp9JtTtCQG6V9TzheCmAsQAvD-BwE&gclidsrc=aw.ds
	Plastic Mosuotrap (6-Pack)	1	pkg	\$ 20.00	\$ 20.00	https://www.amazon.ca/instantly-indoor-outdoor-reusable-plastic/dp/B07WS948GV/ref=sr_1_1?crid=30APX3AZEMT8I&keywords=Mouse%2BTrap%2BTha%2BWork%2BInstantly%2BIndoor%2BOutdoor%2BOnly%2B1&qid=1700989839&s=lawn-garden&srefix=mouse%2Btrap%2Bthat%2Bwork%2BInstantly%2BIndoor%2BOutdoor%2BOnly%2B1%2B1&th=1
	Total Cost				\$ 27.05	

1.5 Aesthetics discussion

The TBWC incorporated several components that the team believed best capture the essence of the Netherlands. A crucial component for any good symbolic representation of a country is the national flag. The Netherlands flag was mounted onto the high walls of the chassis shell so it can be clearly seen. Leading to the shell being painted red, white, and blue for no one to mistake the country we are representing and to show the bold character of the Netherlands people. In addition, we added red, white, and blue flowers alongside the shell. Netherlands is a hotspot known for the endless fields of colorful blooming flowers. It is a major attraction for tourists coming from all around the world as the flower fields enriches the landmark with vibrant spring colors that it has become an important business event during the spring blooming.

1.6 Design Process Summary

Overall, the TBWC design project was a long 7-week process, from the initial troubleshooting of trying to get the chassis to move all the way to creating a launching mechanism that was able to score into the net. Creating a launching mechanism that was completely reliant on the movement of the chassis was one of the most difficult parts of the design. The group chose to utilize a mousetrap that required a string to operate. The back axle pulley would pull the string, creating tension. When the tension reached a certain point, the mousetrap's trigger would release, launching the ball. Throughout the TBWC's development, several problems surfaced. For example, the original wooden mousetrap broke during the construction of the launching mechanism, necessitating the purchase of a more expensive plastic mousetrap alternative in its place. Little things like a wire coming loose from the breadboard would need the team to disassemble the entire TBWC to find the missing wire. This would take several minutes and become an extremely laborious operation as we got closer to finishing our product. Each subsystem had several revisions and many improvements throughout the entire process that would optimize performance, weight, and cost. In effect certain parts would be substituted with others or completely changed to complete the goal at hand. Ultimately was a collective effort of many discussions and tests to create the final product as seen on the performance day.

2. TBWC Performance Summary

Our approach for the final performance was to complete tasks and to attain perfect scores on all the performance tests. The group put countless hours into designing, troubleshooting, and narrowing down each element to give us the best results. On the contest day, the TBWC performed almost perfectly until it came down to the Free Kick portion, where problems began to occur. The first shot was perfect, cleared the wall easily and gracefully fell into the centre of net, however, the second shot did not go as planned and the string responsible for the launching mechanism came undone. Nevertheless, the team came back together and quickly adjusted the string in time for the third attempt, although the ball did not go in, it proved the launching mechanism was still effective and

the code worked as intended. Other than the free kick portion, the TBWC passed every other portion of the performance test nearly perfectly or with a perfect score, getting under a 9 second run for the Ramp dribble consecutively, weighing less than 1200g and not tipping during the safety tests. Overall, the team was satisfied with the TBWC results since it completed tasks to our expectations, and we all felt a sense of relief by the performance that we worked hard to achieve.

Item	Performance	Notes
Ramp Dribble Course Time(s)	8.43s, 8.58s	- the TBWC performed consistently under 9 seconds -slightly slower than expected -completed course without a problem arising
Free Kick Goal Scoring (Outcomes, Points)	Attempt 1: 10 Attempt 2: 3 Attempt 3: 4 Total: 20	-first shot was performed perfectly; ball went over the wall and into the net -the ball did not release during the second shot as the string responsible for releasing the mousetrap came undone -the ball slightly missed the net but made it over the wall -quick troubleshooting after the second shot to ensure the mechanism worked on the third attempt
Mass Measured (g)	1138g	-under the desired 1200g resulting in a perfect score
Aesthetics	8.3	- Needed improvements for branding/appeal -every member provided insight of the desired visual product -Expression was done very well
Safety	Pass	-showed good integrity during the tipping tests -was stable on the ramp and no indication of tipping

		-Teddy bear stayed in place -Required wood trim to aid in stability
Calculated Values		
Mass Calculated(g)	963g	- The calculated mass didn't include all of the nuts and screws and miscellaneous items
Center of Mass Location (x, y, z in cm)	(7.75, 6.23, 7.32)	- The center of mass is near the rear of the wheelchair where most of the weight lies.
Greenhouse Gas Emission (g CO ₂)	2720g	
Water Cyanide Contamination (g CN)	0.34g	
Battery Consumption (Whole Development)	1 Pack (8 batteries)	-Utilized the batteries provided and effectively only used during practice runs and troubleshooting

3. Reflection

3.1 Dribble Ramp (Speed)

The TBWC performed about as we expected for the ramp dribble. The TBWC produced excellent results, consistently finishing within the 8–9 second margin and with only a 0.15 second gap between the two runs. Although originally coded to run approximately under 8 seconds, the ramp clearly created minor implications in the speed expectation of completing the course. Our theorized time for the course was roughly 8 seconds since without the ramp the TBWC went from line A to line C with an average of a little over 7 seconds. Since the chassis was back heavy, we estimated that it would take an additional 0.3 seconds to move up from Line A and 0.5 seconds to go up in reverse from Line C, theoretically making it inside the 8 second margin. It was clear there were no issues with the code as the adjustments made for the ramp created no implications on the speed, since it stopped perfectly at Line C and then reversed back up the ramp. A possible factory reducing speed that was evident from the start of the design process were the tires on the wheels. They were very loose and had lots of empty space in-between them create lots of

drag to the rolling mechanism. After switching to three wheels, we saw lots of improvement in movement as there was less drag contributing to the rolling aspect. Additionally finding a tight rubber band that fit within the pulleys was essential to optimize the spinning from the motor and the speed performed. Nevertheless, the TBWC did a fantastic job during the ramp dribbling because, aside from some new wheels, no significant modifications are needed to achieve the best results.

3.2 Free Kick Shooting (Accuracy, Consistency)

The team's free kick shooting began flawlessly. The TBWC stopped just in front of the wall, waiting for the chance to show off its full potential before releasing the ball and scoring a beautiful goal on its first attempt. However, things took a turn for the worse on the second shot. With team's completing their second attempt, we began to rush while recalibrating the string for the launcher. For the launching mechanism to function, the string needs to be carefully wound around the mousetrap's base and trigger. This requires patience and great attention to detail. We did not have enough time to make sure the launcher was calibrated correctly because the teams were moving faster than we had anticipated. Nevertheless, we took great care to assemble the launcher correctly after the second shot failed to release the ball. On the third try, the ball was fired, but it landed inches shy of the net. It was clear to the team that there were no issues with the launching mechanism at all, but the string became a little tedious to deal with. Overall, the code ran as expected for the TBWC, it would stop several inches away from the wall, while stopped it would have generated enough distance of string to activate the launcher. Going forward, the team needs to address the challenge of recalibrating the string without the necessity of adjustments after each shot. One potential solution involves substituting the existing string with fishing line, which offers enhanced durability and superior elasticity, maintaining its shape effectively. While previous designs incorporated an extra pulley to tension the string, this approach introduced significant drag on the rear wheel axle, resulting in decreased speed, or worse, impeding the chassis from rolling altogether. Consequently, the team had to abandon this idea entirely.

3.3 Mass

A primary objective for the team was to minimize mass, driven by the belief that reducing weight would elevate the performance of the TBWC. Collectively, we reached a unanimous decision that an extra motor was unnecessary for the development of the launching mechanism. Despite the potential for increased workload, prioritizing weight reduction was a consensus among the team since the prototype development stages. The desired mass for the following competition was less than 1200 g, with our TBWC weighing a total of 1138 g. We successfully achieved our goal, and the TBWC demonstrated full functionality, operating at peak performance without the need for an additional motor.

3.4 Aesthetics

The team did well in the aesthetic aspect of the TBWC, with some areas identified for improvement. Overall, each category scored at least an average of 4, which the team was satisfied with. A significant area requiring enhancement was the branding and appeal of the TBWC, as it predominantly featured three colors—red, white, and blue. Apart from the additional flowers, which had symbolic meaning, the shell also acted as a bigger and better flag, which isn't always the most attractive feature of an aesthetic element. In retrospect the team could have added more symbolic representation of Netherlands on the shell since it was primarily a large canvas to attract individuals. Despite this, the remaining three components, craftsmanship, creativity, and expression all performed very well. Expression received the highest score, reflecting the team's dedicated effort in delivering a presentation of exceptional quality. The inclusion of an insightful voiceover and a high-quality video led the team to anticipate that this segment would excel.

3.5 Safety

In terms of the safety portion, the TBWC passed all the test during the final performance testing. During the tipping test, the TBWC showed good integrity and stayed in position without tipping securing the teddy bear. Although for certain positions it required the wood trim to stay in place, it was evident that the TBWC was heavier of the left as of the result of the launcher being on that side. Nonetheless, the teddy bear did not tip on the incline and demonstrated great safety features. Furthermore, the Teddy bear was tightly secured with the chassis and shown no indication of falling out at a point conducting the tests, indicating that the teddy bear holder within the chassis was effective.

3.6 Cost Analysis Table

Date	Material & Part Number	Quantity	Unit	Cost/ Part (\$)	Cost (\$)	Cost Source
2023-10-18	12V/8-AA Battery Pack	1	Indiv.	\$ 5.00	\$ 5.00	TBWC Project Document
	Battery Pack Clip	1	Indiv.	\$ 3.00	\$ 3.00	TBWC Project Document
	AA Batteries	8	Indiv.	\$ 1.00	\$ 8.00	TBWC Project Document
	DC Brushed Motor	1	Indiv.	\$ 10.00	\$ 10.00	TBWC Project Document
	Breadboard	1	Indiv.	\$ 5.00	\$ 5.00	TBWC Project Document
	Solid Core Wire (1 m)	1	Indiv.	\$ 2.00	\$ 2.00	TBWC Project Document
	Arduino UNO	1	Indiv.	\$ 40.00	\$ 40.00	TBWC Project Document
	L6205 Motor Driver Integrated Circuit	1	Indiv.	\$ 12.00	\$ 12.00	TBWC Project Document
	5.6 nF Ceramic capacitor (562)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100uF Electrolytic Capacitor (100uF)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100nF Ceramic Capacitor (104)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100 Ohm Resistor	1	Indiv.	\$ 0.10	\$ 0.10	TBWC Project Document
	10nF Ceramic Capacitor (103)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	1N4148 Diode	2	Indiv.	\$ 0.50	\$ 1.00	TBWC Project Document
	220nF Ceramic Capacitor (224)	1	Indiv.	\$ 0.50	\$ 0.50	TBWC Project Document
	100 kOhm Resistor	1	Indiv.	\$ 0.10	\$ 0.10	TBWC Project Document
	Meccano Part 52 (87 Holes)	1	Part	\$ 8.70	\$ 8.70	TBWC Project Document
	Meccano Part 51E (21 Holes)	2	Part	\$ 2.10	\$ 4.20	TBWC Project Document
	Meccano Part 89 (11 Holes)	2	Part	\$ 1.10	\$ 2.20	TBWC Project Document
	Meccano Part 9F (6 Holes)	2	Part	\$ 0.60	\$ 1.20	TBWC Project Document
	Meccano Part 133C (4 Holes)	6	Part	\$ 0.40	\$ 2.40	TBWC Project Document
	Meccano Part 51F (25 Holes)	1	Part	\$ 2.50	\$ 2.50	TBWC Project Document
	Meccano Part 161 (12 Holes)	2	Part	\$ 1.20	\$ 2.40	TBWC Project Document
	Meccano Part 47 (11 Holes)	1	Part	\$ 1.10	\$ 1.10	TBWC Project Document
	Meccano Part 316B (5 inches)	1	Part	\$ 5.00	\$ 5.00	TBWC Project Document
	Meccano Part 17 (3 inches)	1	Part	\$ 3.00	\$ 3.00	TBWC Project Document
	Meccano Nuts		Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Bolts		Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Spacers		Part	\$ 2.00	\$ 2.00	TBWC Project Document
	Meccano Part 22B3P	1	Part	\$ 1.00	\$ 1.00	TBWC Project Document
	Meccano Part 23B	1	Part	\$ 0.50	\$ 0.50	TBWC Project Document
2023-11-01	Meccano Part 187C	3	Part	\$ 1.00	\$ 3.00	TBWC Project Document
	Meccano Part 142R	3	Part	\$ 1.00	\$ 3.00	TBWC Project Document
	On-Off Switch	1	Indiv.	\$ 3.00	\$ 3.00	TBWC Project Document
2023-11-06	Meccano Part 5 (5 Holes)	1	Part	\$ 0.50	\$ 0.50	TBWC Project Document
	Meccano Part 6A (3 Holes)	1	Part	\$ 0.30	\$ 0.30	TBWC Project Document
	Meccano Part 48A (7 Holes)	1	Part	\$ 0.70	\$ 0.70	TBWC Project Document
2023-11-15	Meccano Part 89 (11 Holes)	3	Part	\$ 1.10	\$ 3.30	TBWC Project Document
	Meccano Part 126 (7 Holes)	2	Part	\$ 0.70	\$ 1.40	TBWC Project Document
	Meccano Part 4 (6 Holes)	2	Part	\$ 0.60	\$ 1.20	TBWC Project Document
	Meccano Part 5 (5 Holes)	1	Part	\$ 0.50	\$ 0.50	TBWC Project Document
	Meccano Part 133C (4 Holes)	1	Part	\$ 0.40	\$ 0.40	TBWC Project Document
	Meccano Part 6A (3 Holes)	2	Part	\$ 0.30	\$ 0.60	TBWC Project Document
2023-11-08	Cardboard Box	70	g	\$ 0.01	\$ 0.70	Recyable material cost- TBWC Project Document
	Roll of String	20	g	\$ 0.20	\$ 0.20	Recyable material cost- TBWC Project Document
			2 oz			
	Red Paint by Craft	1		\$ 1.00	\$ 1.00	https://www.michaels.com/product/acrylic-paint-by-craft-smart-2oz-10694310?michaelsStore=5184&inv=8
	White Paint by Craft	1	2 oz	\$ 1.00	\$ 1.00	https://www.michaels.com/product/acrylic-paint-by-craft-smart-2oz-10694310?michaelsStore=5184&inv=8
	Blue Paint by Craft	1	2 oz	\$ 1.00	\$ 1.00	https://www.michaels.com/product/acrylic-paint-by-craft-smart-2oz-10694310?michaelsStore=5184&inv=8
	Flower Cutouts	15	g	\$ 0.15	\$ 0.15	Recyable material cost- TBWC Project Document
			75 ft.			https://www.michaels.com/product/75ft-gold-natural-brown-twine-by-ashland-10721856?michaelsStore=5184&inv=98
			pkg			
	Plastic Mosusetrap (6-Pack)	1		\$ 20.00	\$ 20.00	https://www.amazon.ca/instantly-Indoor-Outdoor-Reusable-Plastic/dp/B07WS94BGV/ref=sr_1_1?crid=30APX3AZEMT8L&keywords=Mouse%2BTrap%2BThat%2BWork%2BInstantly%2BIndoor%2BOutdoor%2BOnly%2B1&qid=1700989839&s=lawn-garden&srefix=mouse%2Btrap%2Bthat%2Bwork%2BInstantly%2BIndoor%2BOutdoor%2BOnly%2B1%2Clawngarden%2C65&sr=1-1&th=1
Total Cost					\$ 173.85	

3.7 Life Cycle Analysis

Once calculated it was determined that our TBWC design released a total carbon emission of 2720 g, 4.6 g methane, and 0.34g of cyanide.

4. Reflection- Team Dynamics

In terms of expectations and guidelines, the team contract provided a solid foundation for attaining our goals. Throughout the extensive 7-week development period, everyone maintained a respectful demeanor. All suggestions and tactics put forth by each team member were embraced in one way or another, contributing to the collaborative outcome of the project. A significant aspect that the team modified after the prototype meeting was the communication method. Issues arose with MS Teams, such as members receiving late notifications or not receiving them at all. Consequently, the team unanimously decided to transition to Instagram as the primary communication platform for discussions and goal setting, reserving MS Teams for formal reminders about upcoming meetings and schedules. While most work was still shared on MS Teams, Instagram offered a more immediate messaging system, proving useful for unexpected meeting absences or urgent matters not posted on MS Teams.

Engineering Analysis A- Centre of Mass Determination and Tipping Analysis

A.1 Engineering Analysis A

Object	mass (g)	$\hat{x}_i$ (cm)	$\hat{y}_i$ (cm)	$\hat{z}_i$ (cm)
wheel 1	28.6	2.2	1.1	2.3
2	29.1	2.2	13.5	2.3
3	28.2	25.3	6.8	2.3
Breadboard	82.1	13.9	5.5	2.7
- Arduino	28.1	17.4	5.1	2.8
Battery	205.1	9.8	5.8	8.7
Motor	198.2	5.6	5.9	8.1
Mouse Trap	42.2	-2	7.8	12.2
Large Plate	92.4	13.8	5.0	5.2
Curved 1	9.9	7.8	2.8	4.6
2	9.7	7.8	9.6	4.6
3	10.1	13.1	11.2	8.3
Motor Plate	16.3	5.4	5.7	8.1
Motor	130g	6.1	6.0	8.1
Holder				
Shooter	71	-2	8.1	11.4
Sum	963.0g			

$m \cdot \hat{x}$	$m \cdot \hat{y}$	$m \cdot \hat{z}$	$Com \hat{x} = \frac{\sum m \cdot \hat{x}}{m}$
77.2	31.5	65.78	$= 7.75cm$
78.57	392.9	66.9	$Com \hat{y} = \frac{\sum m \cdot \hat{y}}{m}$
712.46	191.8	65	$= 6.23cm$
1124.77	451.55	221.67	$Com \hat{z} = \frac{\sum m \cdot \hat{z}}{m}$
482.9	143.3	78.7	$= 7.32cm$
2009.9	1190	1784.37	$Com = (7.75, 6.23, 7.32)$
1110	1129	1605.42	
-84.4	330	515	
1275	462	481	
77	27.7	45.5	
75.3	93.12	44.6	
71.8	113	84	
88	93	132	
800	780	1057	
		810	

The Hand calculations and the excel sheet calculations were similar and had little difference which come down to human error. On the day of the challenge our wheelchair weighed in at 1138g and our calculations showed around 980g which is less because in the calculations we couldn't include all the nuts and bolts and miscellaneous items which will get us to the weight. Also, on the day of the test our wheelchair didn't tip and was balanced in all the ways tested which agree with our calculations.

angle Calculation			
Left	Z-x	Z-y	
	$\theta = 43.4^\circ$	Front	$\theta = 49.6^\circ$
Right	Z-x		
	$\tan \theta = \frac{7.32}{7.75}$	Back	$\tan \theta = \frac{7.32}{6.23}$
	$\theta = 42.4^\circ$		$\theta = 49.6^\circ$

Engineering Analysis B- Life Cycle Impacts Analysis

B.1 Engineering Analysis B

		CO ₂	methane	Cyanide
Wheel 1, 2, 3				
Synthetic rubber	36g	$4 \times 10^3 = 0.144$		
Breadboard PCB				
PolyPropylene	82.1g	$1.5 \times 10^3 = 0.123$	1.5×10^{-1} $= 0.00123$	1.4×10^{-8} $= 1.15 \times 10^{-12}$
Arduino	28.1g	$6 \times 10^3 = 0.17$	9.3	0
wood			$= 2.6 \times 10^{-4}$	-0
Battery	205.1g	$2.6 \times 10^3 = 0.533$	1.4	6.3×10^{-1}
Steel			$= 2.9 \times 10^{-4}$	$= 1.3 \times 10^{-9}$
Mouse trap	42.2g	$1.5 \times 10^3 = 0.0633$	1.5×10^{-1}	1.4×10^{-8}
PolyPropylene			$= 6.3 \times 10^{-4}$	$= 5.9 \times 10^{-13}$
Motor	198.2g	$2.6 \times 10^3 = 0.52$	1.4	6.3×10^{-1}
Steel			$= 2.8 \times 10^{-4}$	$= 1.2 \times 10^{-4}$
Large Plate	92.4g	$4.1 \times 10^3 = 0.379$	8.9	2×10^{-4}
Stainless Steel			$= 8.2 \times 10^{-4}$	$= 1.8 \times 10^{-8}$
Curved 1 2 3	29.7g	$4.1 \times 10^3 = 0.121$	8.9	2×10^{-4}
Stainless Steel			$= 2.6 \times 10^{-4}$	$= 5 \times 10^{-9}$
Motor Plate + Holder	146.3g	$2.6 \times 10^3 = 0.38$	1.4	6.3×10^{-1}
Steel			$= 2.04 \times 10^{-4}$	$= 9.2 \times 10^{-5}$
Shooter	71g	$4.1 \times 10^3 = 0.2911$	8.9	2×10^{-4}
Stainless			$= 6.3 \times 10^{-4}$	$= 1.4 \times 10^{-8}$
Sum		2.72kg $= 2720g$	-0.0046 kg $= 4.6g$	-0.00034 kg $= 0.34g$

The total CO₂ emissions in our wheelchair is 2720 g, the total methane emissions in our wheelchair were 4.6 g, and our total amount of cyanide was 0.34 g.

Engineering Analysis C- Arduino Code

C.1 Engineering Analysis C

```
1 // define pin names//
2 #define enA 3
3 #define in1A 4
4 #define in2A 5
5
6
7 void setup() {
8 //setup for first part of test
9
10 // initialize serial communication
11 Serial.begin(9600);
12
13 // set the pin modes as outputs (signals output from Arduino to breadboard)
14 pinMode(enA, OUTPUT);
15 pinMode(in1A, OUTPUT);
16 pinMode(in2A, OUTPUT);
17
18 // enable motor A by setting enableA high permanently
19 digitalWrite(enA, HIGH);
20
21 }
22
23 void loop() {
24 // put your main code here, to run repeatedly:
25
26 // spin the motor in one direction
27 digitalWrite(in1A, HIGH);
28 digitalWrite(in2A, LOW);
29 Serial.println("Going towards black line"); // print status message to serial monitor
30
31 delay(3800); // run for time t
32
33 // stop the motor
34 digitalWrite(in1A, LOW);
35 digitalWrite(in2A, LOW);
36 Serial.println("Stopped"); // print status message to serial monitor
37
38 delay(500); // wait for 1500ms
39
40 // spin the motor in one direction
41 digitalWrite(in1A, LOW);
42 digitalWrite(in2A, HIGH);
43 Serial.println("Going towards black line"); // print status message to serial monitor
44
45 delay(4800); // run for time t
46
47
48 exit(0); // vehicle travel finished
49
50 }
```

For our first test we used this code to have our wheelchair complete the course back and forth. This programmed the Arduino to allow the motors to spin and in the clockwise

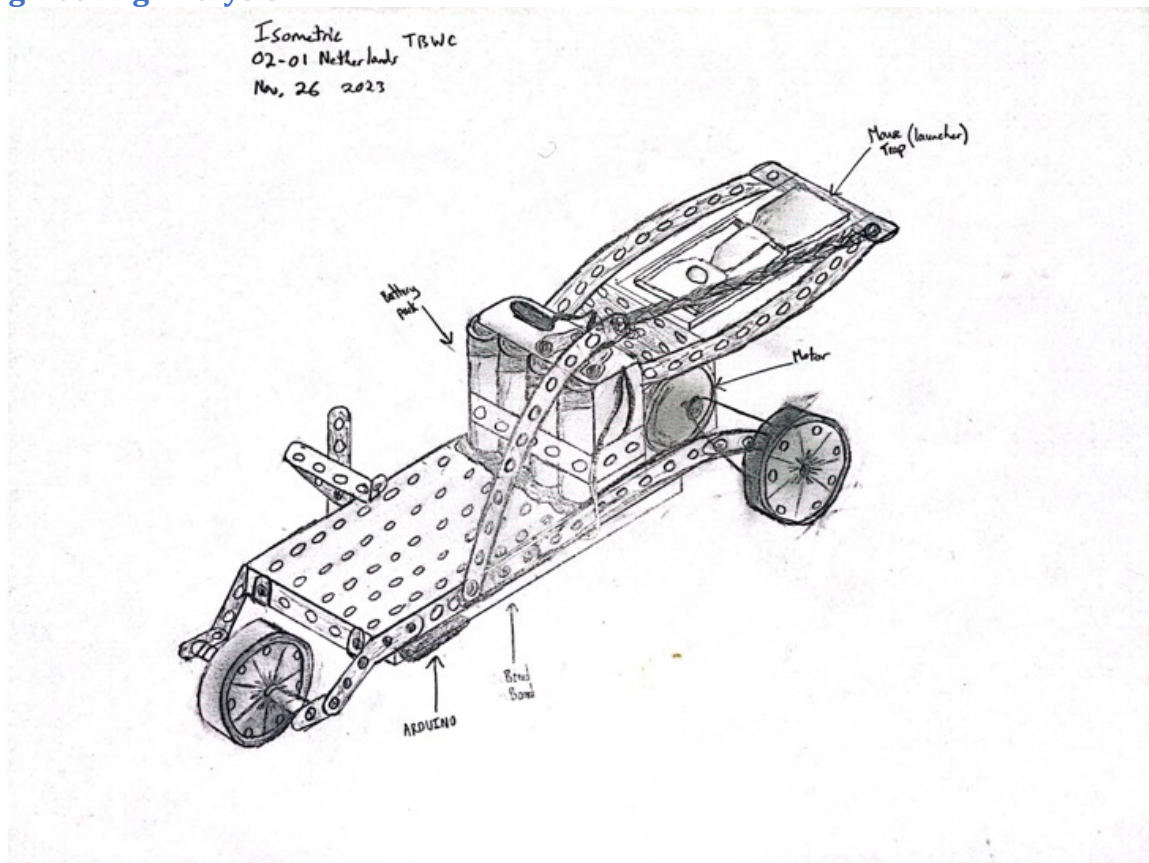
direction to move it forward for 3.8 seconds and then once it reaches the black line it would stop for 0.5 seconds and continue back to the starting line for 4.8 seconds.

```
1 // define pin names//
2 #define enA 3
3 #define in1A 4
4 #define in2A 5
5
6
7 void setup() {
8 //setup for first part of test
9
10 // initialize serial communication
11 Serial.begin(9600);
12
13 // set the pin modes as outputs (signals output from Arduino to breadboard)
14 pinMode(enA, OUTPUT);
15 pinMode(in1A, OUTPUT);
16 pinMode(in2A, OUTPUT);
17
18 // enable motor A by setting enableA high permanently
19 digitalWrite(enA, HIGH);
20
21 }
22
23 void loop() {
24 // put your main code here, to run repeatedly:
25
26 // spin the motor in one direction
27 digitalWrite(in1A, HIGH);
28 digitalWrite(in2A, LOW);
29 Serial.println("Going towards black line"); // print status message to serial monitor
30
31 delay(2200); // run for time t
32
33 // stop the motor
34 digitalWrite(in1A, LOW);
35 digitalWrite(in2A, LOW);
36 Serial.println("Stopped"); // print status message to serial monitor
37
38 delay(1000); // wait for 1500ms
39
40
41 exit(0); // vehicle travel finished
42
43 }
```

For the second test we had to complete we used the blueprints of the first code to our second code as well. We had to move forward for 2.2 seconds and then within that time the string on our wheelchair would wind up and activate the mousetrap to launch the ball. Then after the time is complete since the task is finished the motor stops running and then exits the code.

Engineering Analysis D- Isometric Drawing of Final TBWC

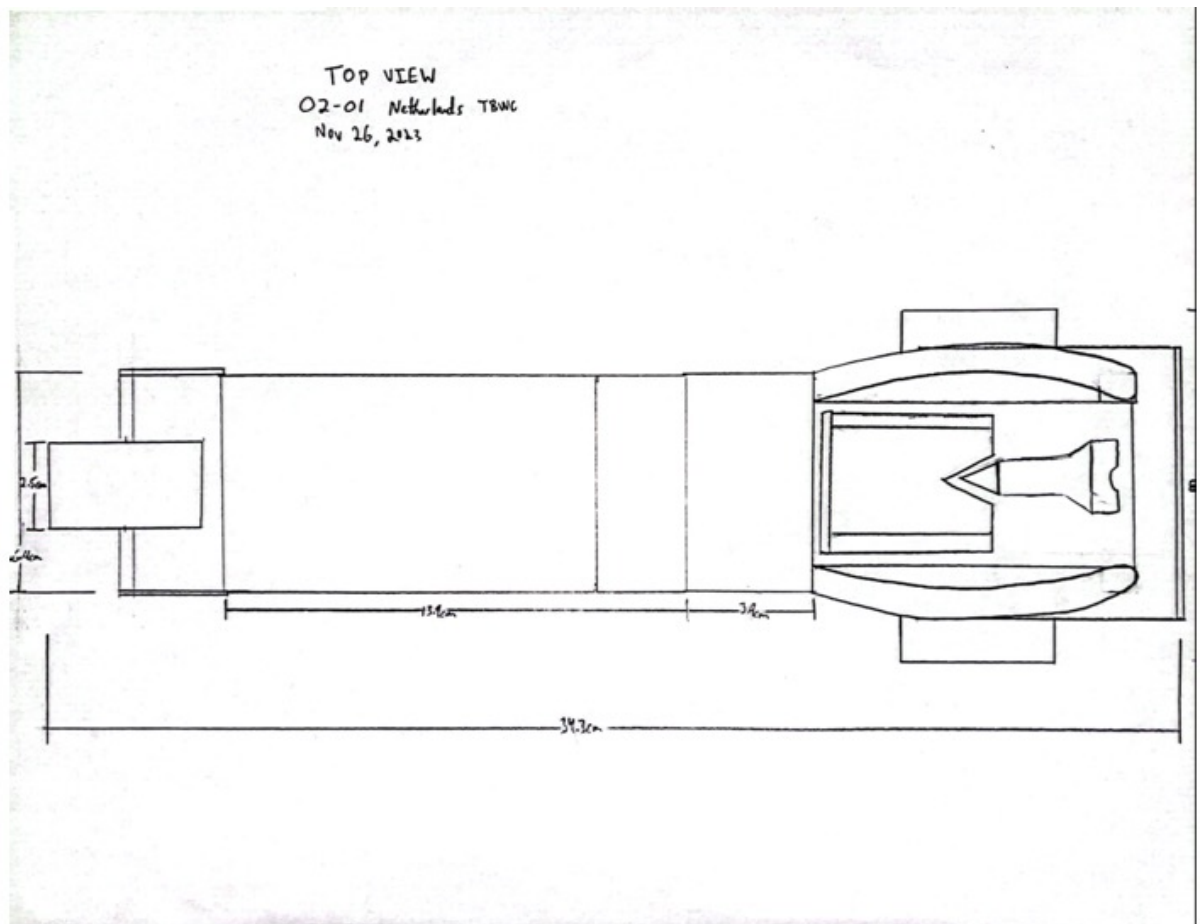
D.1 Engineering Analysis D



Scanned with CamScanner

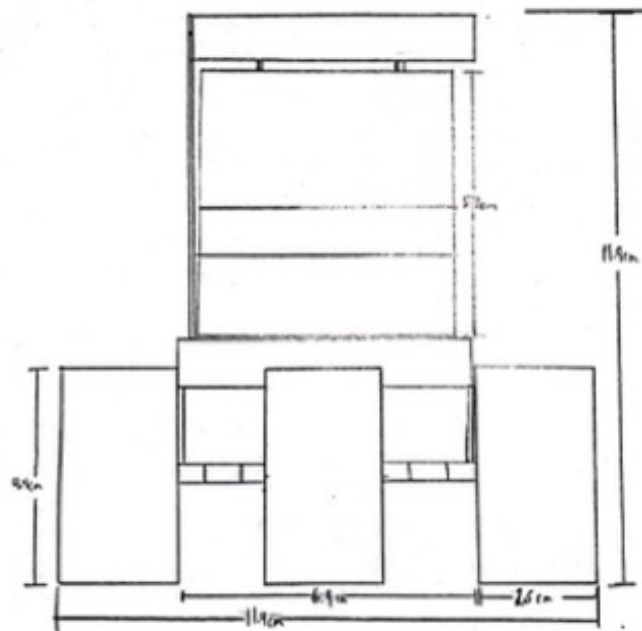
Engineering Analysis E- Orthographic Projections of Final TBWC

E.1 Engineering Analysis E

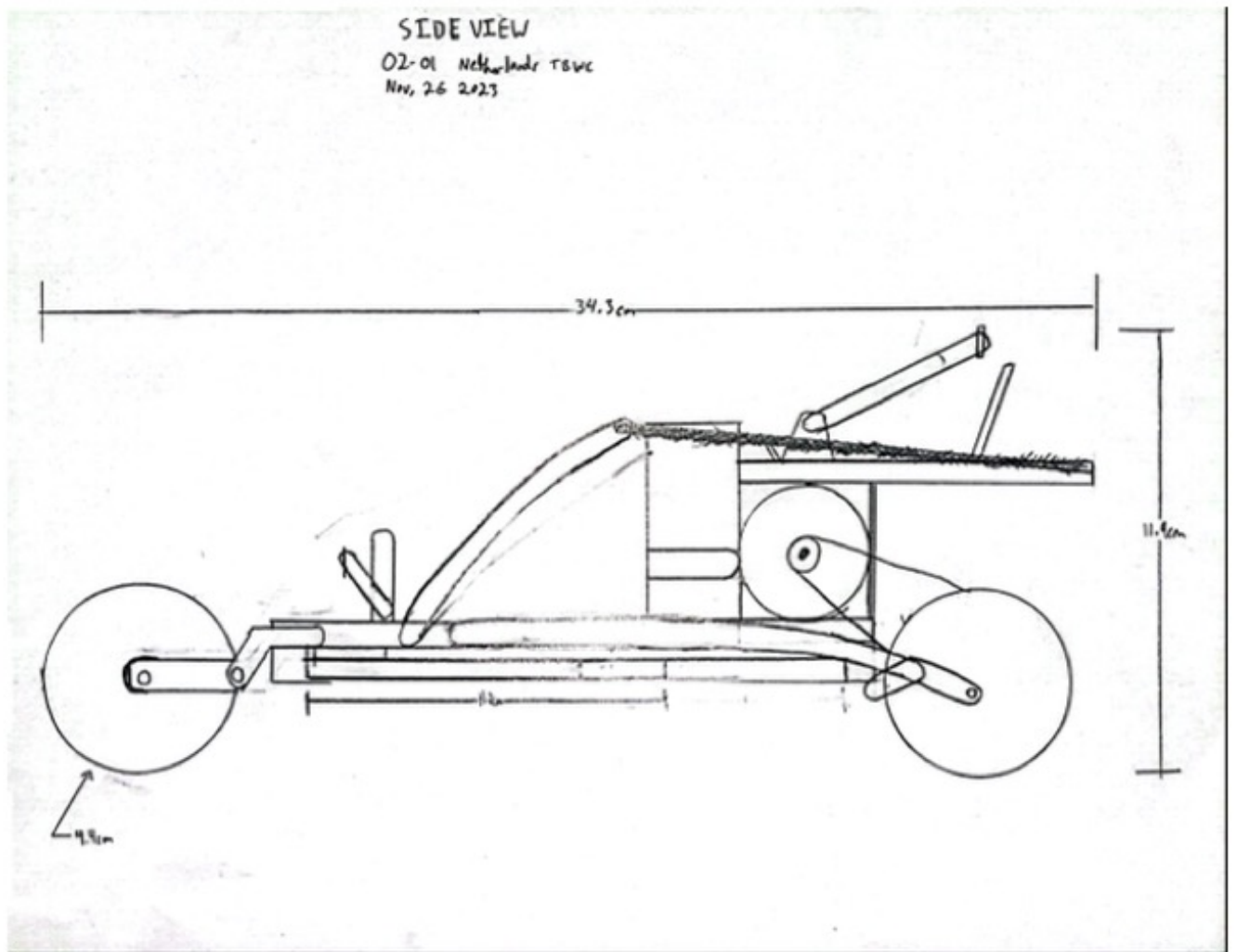


Scanned with CamScanner

FRONT VIEW
02-01 Nthwledr TBWC
Nov 26, 2023



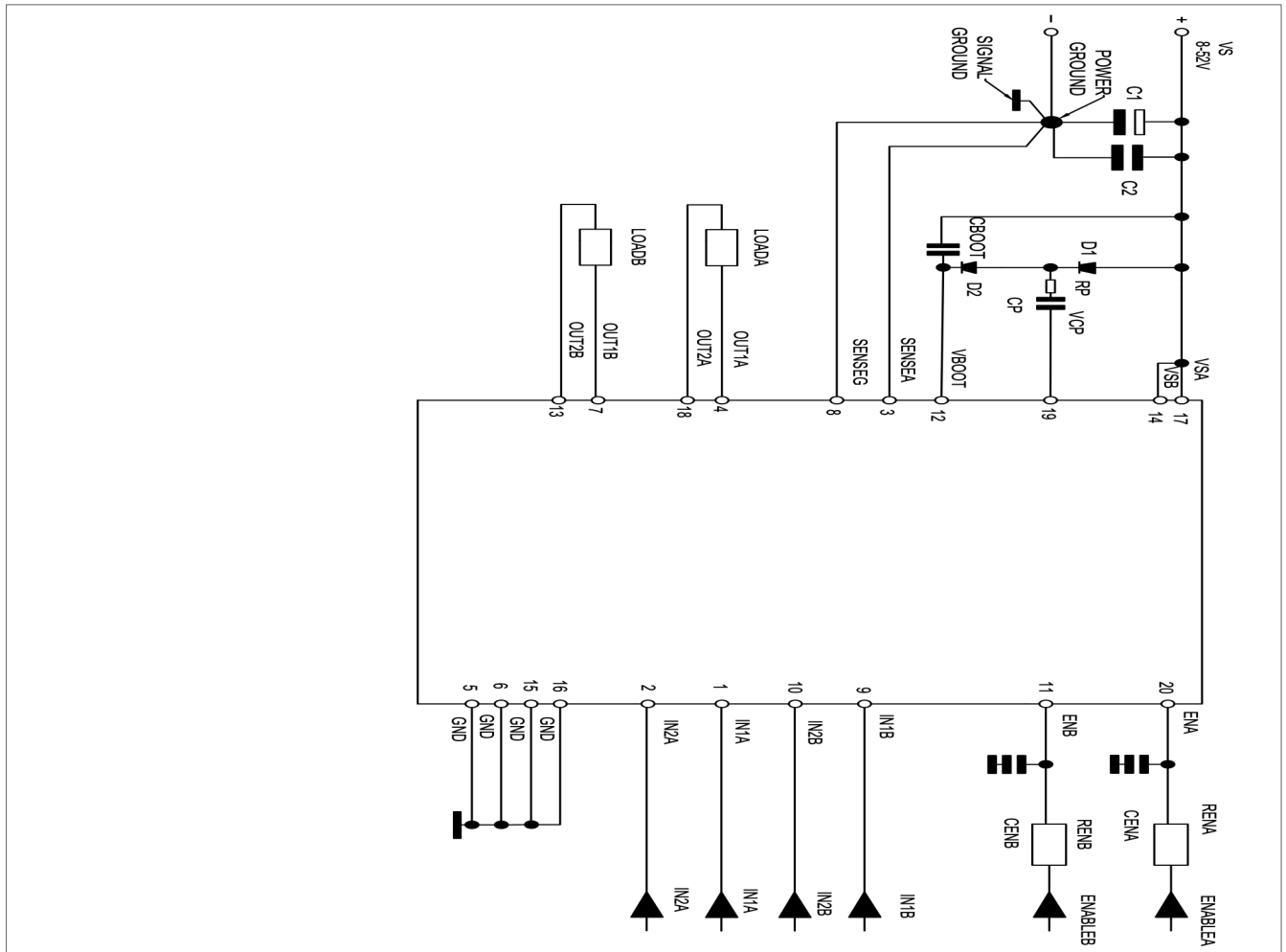
Scanned with CamScanner



Scanned with CamScanner

Engineering Analysis F- TBWC Electrical Circuit

F.1 Engineering Analysis F



Appendix G- TBWC Aesthetic Assessment Rubric and Grading



School of Engineering
Engineering & Design I (ENGG*1100)
TBWC Aesthetic Assessment Rubric and Grading – F23

Aesthetic Assessment for Team **02-01**

Score is weighted 40% Students + 60% Faculty plus GTA

Criteria*	Assessments		Weighted Score (0.4*Student+0.6*Faculty/GTA)
	Student	Faculty/GTA	
Craftsmanship (5)	4.0	4.0	4.0
Creativity (5)	4.1	4.3	4.3
Branding/Appeal (5)	4.2	3.7	3.9
Expression (5)	4.1	4.7	4.4
Total Rubric Score (20)			16.6
TBWC Aesthetic Score (10)			8.3

* TBWC Aesthetic Assessment Rubric on back of page.



Appendix H- TBWC Week 7 Revised Team Contract



School of Engineering
Engineering & Design I (ENGG*1100)

Team Contract Revised– Team TBWC- S02 -1

1. I will communicate with team members through MS teams, Instagram group messages or via SMS group messages. Instagram group messages or SMS messages will be the most convenient way of communication to let team members know of unexpected absences to meetings and will be used to notify the team of urgent meetings. Any ideas/plans for designs, and majority of communication amongst the team will remain on MS teams. Reminders will be posted on MS teams and via Instagram group messages to notify members of meetings.
2. I will attend each meeting, meetings will be scheduled ahead of time by the team leader on MS teams, which will include the location(s) and time for the meeting, everyone is expected to be at meetings.
3. Our team will meet at least once a week for at least 1 hour outside the lab meetings or project work, to review and contribute to the project's overall status.
4. A project schedule will be created to set up goals and expectations for each meeting. In addition, a team member (alternate) will provide the results/ideas/plans gathered from each meeting on a shared document uploaded to MS teams.
5. Each member is expected to treat one another with decency and professionalism. Members who engage in harassment, rudeness, or bullying will not be tolerated; members will be warned, if behavior persists, the TA will be informed.
6. Team members will distribute all tasks equally. Members will be given tasks they are comfortable with and where they can apply their skills to improve the project.
7. A team member will submit each task during the design project to the course link Dropbox before the deadline.
8. All the work completed or ideas I share will be original.